# Linear Regression

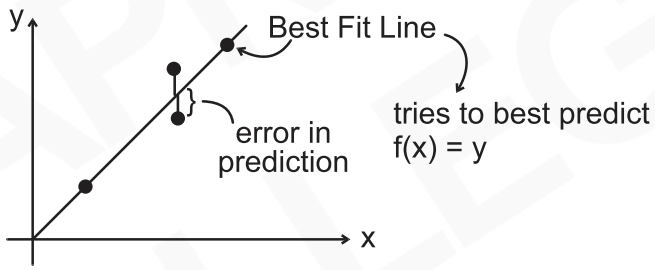
## Intuition & Logic

Linear Regression is a supervised ML algorithm for regression problems.

LR models the relationship between a dependent variable (output) and one or more independent variables (inputs) by fitting the best straight line (or plane/hyperplane) to the data.

In the simplest form the linear regression can be understood by taking an example of single input feature (x) & output (y).

LR is a regression algorithm that tries to predict the relation of x & y in the form of a BEST FIT LINE.




eqⁿ of straight line ⇒ y = mx + c
which gives us our LR hypothesis function eqⁿ :-

$$y = c + mx$$

$$h_\theta(x) = \theta_0 + \theta_1 x$$

- $h_\theta(x)$: hypothesis function or our model
- $\theta_0$: intercept (bias)
- $\theta_1$: slope (or coefficient of x) or weights

When we have multiple independent features, it is called multiple linear regression:-

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

**Goal:** Best estimate all $\theta_i$ so that best fit line (or plane) best predicts y.

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## How to find this Best fit line?

We find it by trying to minimize the **Cost function**.

**Cost function** is a function that measures how far the predicted values $(\hat{y})$ are from actual values $(y)$.

Most common CF used for LR is Mean Squared Error (MSE) :-

$$J(\theta) = \frac{1}{2.m}\sum_{i=1}^{m}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

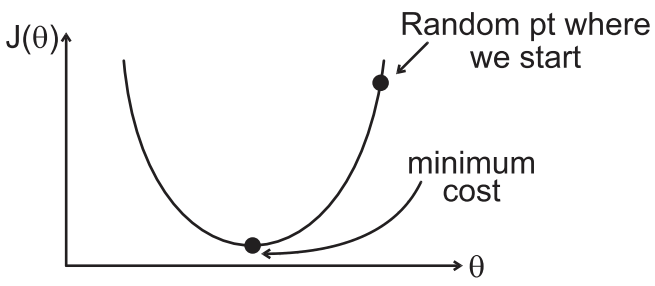
- Cost function
- Divided by 2 for calculation simplicity after derivation
- m are total samples in our dataset
- $\hat{y}$ (prediction)
- $(\hat{y} - y)$ error
- y (actual)

Now that we know we have to minimize cost function, we do so by using a technique called **Gradient Descent**.

GD is an iterative technique that iteratively updates $\theta_0$ & $\theta_1$ until the MSE (or $J(\theta)$) reaches its lowest value.

## How does Gradient Descent for LR work?

GD is an optimization technique used to train our LR model by minimizing prediction error.
To understand GD, let's plot $\theta$ & $J(\theta)$

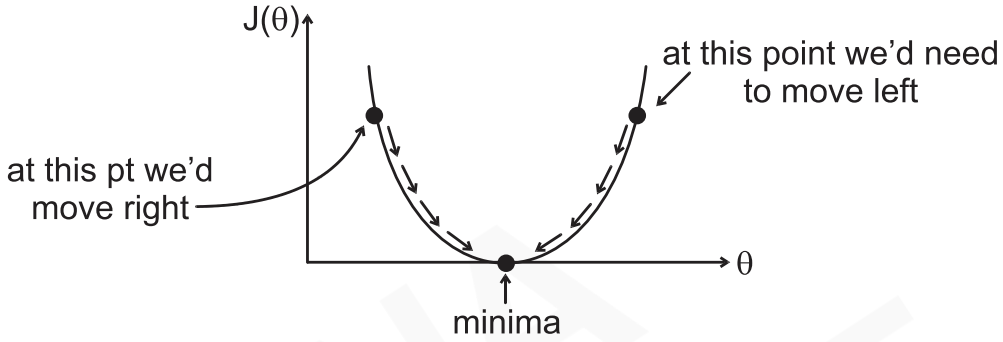



we get such a curve. Minimum cost is at the Global Minima. We iteratively try to converge to this value using GD.

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How? Using these steps:-

1. Start with random value of $\theta_0$ & $\theta_1$.
2. Calculate the error between j & y using MSE i.e. $J(\theta)$
3. Compute Gradient i.e. derivative of cost function. Why? Because it is essentially the slope which will point in the direction of the steepest increase




so for $J(\theta_0, \theta_1) = \frac{1}{2.m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$

we find $\frac{\partial J(\theta)}{\partial \theta_0}$ & $\frac{\partial J(\theta)}{\partial \theta_1}$

**Extra**

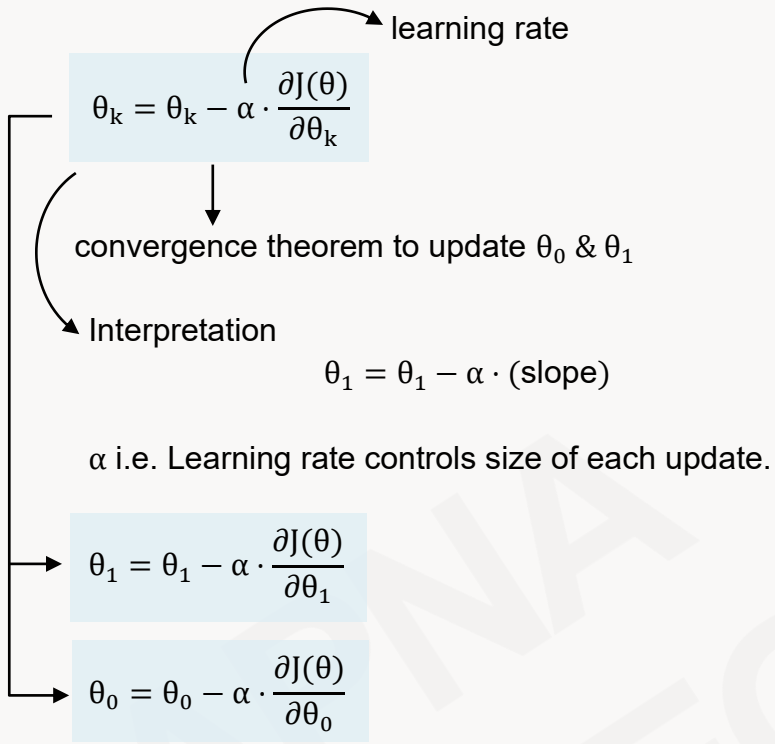
if $J(\theta_0, \theta_1) = \frac{1}{2m}\sum_{i=1}^{m}\left((\theta_0 + \theta_1 x_i) - y_i\right)^2$

so $\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{m}\cdot\sum_{i=1}^{m}(\underbrace{(\theta_0 + \theta_1 x_i)}_{\hat{y}_i} - y_i)$

$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{m}\sum_{i=1}^{m} x_i(\underbrace{(\theta_0 + \theta_1 x_i)}_{\hat{y}_i} - y_i)$

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4. Update Parameters $\theta_0$ & $\theta_1$ to reduce the error




5. Keep repeating this process (Steps 2 to 4) until error stops decreasing significantly.

**Special Note** - For simple linear regression we can use formulas like Normal Equation i.e.

$$\theta = (X^T X)^{-1}.(X^T y)$$

to find parameters directly (without Gradient Descent). So sklearn directly estimates the coefficients using the Ordinary least squares (OLS) method.

However for large datasets or high-dimensional data these methods become computationally expensive that's why we need Gradient Descent.

Also, in polynomial regression, the cost function becomes highly complex and non-linear, so analytical solutions are not available. That's where gradient descent plays an important role.

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# Linear Regression Assumptions

Every ML model has some assumptions which are foundational conditions that must hold true for the model's results to be reliable, accurate & generalizable.

Linear regression relies on several key assumptions, often remembered by the acronym **LINE** (with some extensions):

- **L - Linearity:** The relationship between the features and target.
- **I - Independence:** Observations are independent of each other.
- **N - Normality:** The error(residuals) follows a normal distribution.
- **E - Equal Variance (Homoscedasticity):** The error term has a constant variance.
- **Multicollinearity:** There is no multicollinearity between the features.

We can breakdown these into different categories:

Assumptions about the **residuals**:

- **Normality assumption:** The error terms, ε(i), are normally distributed.
- **Zero mean assumption:** The residuals have a mean value of zero.
- **Constant variance assumption:** The residual terms have the same (but unknown) value of variance, σ^2. This assumption is also called the assumption of homogeneity or homoscedasticity.
- **Independent error assumption:** The residual terms are independent of each other, i.e. their pair-wise covariance value is zero.

Assumptions about the **estimators**:

- The independent variables are measured without error.
- There does not exist a linear dependency between the independent variables, i.e. there is no multicollinearity in the data.

*Keep Learning & Keep Exploring!*

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